

MATH FAT

Discrete
mean = $\frac{\sum x}{N}$

Grouped
 $\frac{\sum f_i x_i}{N}$
(or)

$A + \frac{\sum f_i d_i}{N}$

$d_i = x_i - A$

Grouped Continuous
 $\frac{\sum f_i x_i}{N}$ $x_i = \frac{l+u}{2}$

or

$A + \frac{\sum f_i d_i \times c}{N}$

$d = \frac{x_i - A}{c}$

Median
Discrete $\rightarrow$ sort $\rightarrow$ $\frac{(\frac{n}{2}) + (\frac{n}{2} + 1)^{th}}{2}$
even $\rightarrow$ Avg of

Grouped $\rightarrow$ value where $cf \geq N/2$
 $N = \sum f_i$

Continuous $\rightarrow$ $l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$

$l \rightarrow$ lower boundary of median class

$N = \sum f_i$

$cf =$ c.f before median class

$f \rightarrow$ f of median class

$h \rightarrow$ height

Mode
Discrete $\rightarrow$ most repeated

&
Grouped

Continuous

$l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$

$f_1 \rightarrow$ f of modal class

$f_0 \rightarrow$ f of preceding modal

$f_2 \rightarrow$ f, after modal

$l \rightarrow$ lower boundary of modal

$h \rightarrow$ height.

Standard dev

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{N}}$$

$$\text{or} \quad \sqrt{\frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2}$$

$$C.V = \frac{\sigma}{\bar{x}} \times 100$$

and by assumed mean

$$\sigma = \sqrt{\frac{\sum f_i d_i^2}{\sum f_i} - \left(\frac{\sum f_i d_i}{\sum f_i}\right)^2}$$

$$d_i = \frac{m_i - a}{h} \quad \text{for continuous}$$

$$d_i = m_i - a \rightarrow \text{grouped}$$

$$P(A/B) = \frac{P(A)}{P(B)} \cdot P(B/A)$$

$$P(A \cap B) = P(A) \cdot P(B/A)$$

PMF
→

$$\sum p(x) = 1.$$

$$p(x) \geq 0 \quad \forall x$$

CDF

$$\int_{-\infty}^{\infty} f(x) = 1.$$

$$f(x) = \frac{d}{dx} (F(x))$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

$$E(x) = \sum x P(x=x)$$

or

$$\int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$\text{Variance} = E(x^2) - (E(x))^2$$

or

$$E[(x-\mu)^2]$$

Binomial

→

$$P(x=k) = {}^nC_k \cdot p^k \cdot q^{n-k}$$

$$\text{mean} = E(x) = np$$

$$\text{variance} = npq$$

$$\text{s.d} = \sqrt{npq}$$

Poisson:

$$P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$$\lambda = \text{mean} = np$$

$$\text{Var}(X) = E(X) = \text{mean} = \lambda$$

Normal distribution:

$$P(X=x) = \frac{1}{(\sqrt{2\pi})\sigma} \cdot e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

for standard normal

$$\mu = 0, \sigma = 1$$

$$\text{i.e. } P(X=x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

for z-value

$$z = \left(\frac{x-\mu}{\sigma} \right)$$

$$\text{i.e. } P(X=z) = \frac{1}{(\sqrt{2\pi})\sigma} e^{-z^2/2}$$

Karl Pearson's

$$r = \frac{\text{Cov}(x, y)}{\sqrt{\text{Cov}(x, x) * \text{Cov}(y, y)}}$$

$$\text{Cov}(x, y) = \text{mean}(x * y) - \text{mean}(x) * \text{mean}(y)$$

Regression =

$$y = a + bx$$

$$b = \frac{\text{Cov}(x, y)}{\text{Cov}(x, x)}$$

$$(a = \bar{y} - b\bar{x})$$

$$r^2 = b_{xy} \cdot b_{yx}$$

y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$b_{yx} = r \left(\frac{\sigma_y}{\sigma_x} \right)$$

$$\text{and } b_{xy} = r \left(\frac{\sigma_x}{\sigma_y} \right)$$

Multiregression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

$$E_y = n\beta_0 + \beta_1 E x_1 + \beta_2 E x_2$$

$$E_{x_1 y} = \beta_0 E x_1 + \beta_1 E x_1 \cdot x_1 + \beta_2 E x_1 \cdot x_2$$

$$E_{x_2 y} = \beta_0 E x_2 + \beta_1 E x_2 \cdot x_1 + \beta_2 E x_2 \cdot x_2$$

Solve to get $\beta_0, \beta_1, \beta_2$

Spearman's Rank

$$r = 1 - 6 \left(\frac{\sum D_i^2 + \frac{1}{12}(m^3 - m)}{n(n^2 - 1)} \right)$$

$$D_i \rightarrow R_i - R_j$$

if repeated ranks
assign avg.

P-Value test:

$$H_0: p = p_0$$

$$H_1: p \neq p_0 \text{ (or) } p > p_0 \text{ (or) } p < p_0$$

$$p < p_0 \rightarrow P = P(X \leq x) \leq \alpha \rightarrow \text{reject } H_0$$

$$p > p_0 \rightarrow P = P(X \geq x) \leq \alpha \rightarrow \text{reject } H_0$$

$$p \neq p_0 \rightarrow P = 2P(X \geq x) \leq \alpha \rightarrow \text{reject } H_0$$

$$P(X = x) = {}^n C_x \cdot p^x \cdot q^{n-x}$$

$$\text{or } z = \frac{x - \mu}{\sigma} = \left(\frac{x - np_0}{\sqrt{np_0 q_0}} \right)$$

2 proposition

$$H_0: p_1 = p_2$$

$$p_1 = \frac{x_1}{n_1}, p_2 = \frac{x_2}{n_2}, \hat{p} = \frac{x_1 + x_2}{n_1 + n_2}, \hat{q} = 1 - \hat{p}$$

$$H_1: p_1 < p_2: P(Z \leq z) \leq \alpha, \text{ reject } H_0$$

$$p_1 > p_2: P(Z \geq z) \leq \alpha, \text{ reject } H_0$$

$$p_1 \neq p_2: 2P(Z \geq |z|) \leq \alpha, \text{ reject } H_0$$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Variance known:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$\mu < \mu_0: P(Z \leq z) \leq \alpha, \text{ reject } H_0$$

$$\mu > \mu_0: P(Z \geq z) \leq \alpha, \text{ reject } H_0$$

$$\mu \neq \mu_0: 2P(Z \geq |z|) \leq \alpha, \text{ reject } H_0$$

for 2 means

$$Z = \frac{(\bar{x}_1 - \bar{x}_2) - (d_0)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$d_0 = \mu_1 - \mu_2$$

$$\text{If } |Z| > (z_\alpha \text{ or } z_{\alpha/2}) \\ \text{reject } H_0.$$

5%	1%	
z_{α} +1.645	+2.33	right
z_{α} -1.645	-2.33	left
$z_{\alpha/2}$ +1.96	+2.58	2-tailed

t-test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$s_p = \sqrt{\frac{\sum (x_1 - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}}$$

$$df = n_1 + n_2 - 2$$

or

$$df = n - 1 \rightarrow \text{for single}$$

If

~~t_{calc}~~

$$|t_{calc}| > t_{table}$$

↳ reject H_0

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

F-test:

$$F_{\text{stat}} = \frac{s_1^2}{s_2^2} \quad \left(\begin{array}{l} \text{Always} \\ s_1^2 > s_2^2 \end{array} \right)$$

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$df = n_1 - 1, n_2 - 1$$

$$F_{\text{stat}} > F_{\text{table}} \\ \text{reject } H_0$$

ANOVA

$$s_t^2 = \frac{\sum x^2}{N} - C$$

$$C = \frac{(\sum x)^2}{N}$$

$$s_b^2 = \frac{(\sum x_1)^2}{N_1} + \frac{(\sum x_2)^2}{N_2} + \dots - C$$

$$s_w^2 = s_t^2 - s_b^2$$

$$F_{\text{stat}} = \frac{s_b^2 / (k-1)}{s_w^2 / (N-k)}$$

$N \rightarrow$ total rows

$k \rightarrow$ total cols

if

$$F_{\text{stat}} > F_{k-1, N-k}$$

reject H_0

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

H_1 : At least one mean is significantly diff.

F-test Anova

Ex:

$$N=21, k=3$$

	GI	GII	GIII
1	3	4	5
2	5	5	5
3	3	3	5
4	1	4	1
5	7	9	7
6	3	5	3
7	6	5	7
	$\Sigma x_1 = 28$	$\Sigma x_2 = 35$	$\Sigma x_3 = 33$

$$S_t^2 = \Sigma x^2 - C$$

$$C = \frac{(\Sigma x)^2}{N}$$

$$S_b^2 = \frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \frac{(\Sigma x_3)^2}{N_3}$$

$$\Sigma x = \Sigma x_1 + \Sigma x_2 + \Sigma x_3 = 96$$

$$C = \frac{(96)^2}{21} = 438.86$$

$$C = 438.86$$

$$\frac{(96)^2}{21} = 438.86$$

So

$$S_t^2 = \Sigma x^2 - C$$

$$\Sigma x^2 = 522$$

$$\Sigma x^2 = 8314$$

$$S_b^2 = \frac{(28)^2}{7} + \frac{(35)^2}{7} + \frac{(33)^2}{7} - C = 442.57 - 438.86 = 3.71$$

$$S_w^2 = S_t^2 - S_b^2 = 83.14 - 3.71 = 79.43$$

$$K-1 = 2$$

$$N-K = 18$$

$$F = \frac{\frac{S_b^2}{K-1}}{\frac{S_w^2}{N-K}} = \frac{\frac{(3.71)^2}{2}}{\frac{(79.43)^2}{21-3}} = 0.420$$

$$F_{tab} = 3.54$$

$$f_{stat} > F_{tab}$$

fail to reject H0

χ^2 -Test

$$\chi^2 = \sum \left(\frac{(O-E)^2}{E} \right)$$

$$df = n - 1$$

$$\chi^2_{calc} > \chi^2_{table}$$

reject H_0

H_0 : variables are independent

(or)

data follows expected distribution

with

contingency table

$$df = (r-1) \times (c-1)$$

$$E_{r,c} = \frac{(\text{Row}) (\text{Col})}{\text{size}}$$

Ex:

		51	52	43	28	10	216
M	32						
F	13	22	33	21	10	6	105
	45	73	85	64	38	16	321

↓ size

$$E_{32,51} = \frac{45 \times 216}{321} = 30.28$$

$$E_{13} = \frac{45 \times 105}{321} = 14.719$$

32	30.28	13	14.719
51	49.121	22	23.878
52	.	33	.
43	.	21	.
28	.	10	.
10	.	6	.

Combine

σ_1, σ_2
to single

col.
calculate
 $(O-E)^2/E$

Non-Parametric

Sign - test.

$$T^+ \rightarrow +ve$$

$$T^- \rightarrow -ve$$

$$T = \min(T^+, T^-)$$

$$T^0 = \text{No. of zeros}$$

(ignore.)

Ex:

6, 5, 1, 2, 2, 5, 2, 5, 3, 7, 4, 7

median = 4.5

+1, +1, -1, -1, -1, +1, +1, +1, -1, +1, +1

$$T^+ = 7$$

$$T^- = 5$$

$$T = \min(7, 5)$$

$$T = 5$$

$$\frac{n(n+1)}{2} = \frac{T^+ + T^-}{2}$$

$$\frac{n(n+1)}{2} = 19$$

$$n(n+1) = 38$$

Wilcoxon Signed Rank test.

compute

$$T^+, T^-$$

$$T = \min(T^+, T^-)$$

for T_{table}

$$n \leq \frac{n(n+1)}{2} = (T^+ + T^-)$$

find n .

if

$$T < T_{table}$$

↳ reject H_0

for large sample

$$Z = \left(\frac{T - \mu_T}{\sigma_T} \right)$$

$$\mu_T = \frac{n(n+1)}{4}$$

$$\sigma_T = \sqrt{\frac{n(n+1)(2n+1)}{24}}$$

$$|Z| > Z_{tab}$$

reject H_0 .

$T^+ \rightarrow$ sum of ranks of +ve

$T^- \rightarrow$ sum of ranks of -ve

Run Test:

$n_1 \rightarrow$ no. of symbols of 1st type

$n_2 \rightarrow$ of 2nd type

$n \rightarrow$ no. of runs.

Ex - + - + - + + + -

$$n = 7$$

$$n_1 = 5$$

$$n_2 = 5$$

$$n_1 + n_2 = 10.$$

$G_L < R < G_U \rightarrow$ Random.

$R \leq G_L \rightarrow$ Too few runs (Clustering)

$R > G_U \rightarrow$ Too many runs
(Oscillation)

For large sample

$$E(R) = \frac{2n_1n_2}{n_1+n_2} + 1$$

$$Var(R) = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1+n_2)^2(n_1+n_2-1)}$$

$$Z = \frac{R - E(R)}{\sqrt{Var(R)}}$$

$|Z| > z_{\alpha/2} \rightarrow$ reject H_0

(20)

for large sample

$$\mu = \frac{2n-1}{3}, \quad \sigma = \sqrt{\frac{16n-29}{90}}$$

$$r = (\mu \pm 1.96\sigma)$$

$$r_L = (\mu - 1.96\sigma)$$

$$r_U = (\mu + 1.96\sigma)$$

Mann-Whitney U-test:

$$U_1 = R_1 - \frac{n_1(n_1+1)}{2}$$

$$U_2 = R_2 - \frac{n_2(n_2+1)}{2}$$

$$U = \min(U_1, U_2)$$

if $U \leq U_{table}$

for large sample

$$\mu_U = \frac{n_1 n_2}{2}; \quad \sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

$$Z = \frac{U - \mu_U}{\sigma_U}$$

$$|Z| > Z_{table}$$

↪ reject H_0